

# Ruangguru wins trust of teachers and tutors in digital education services with Google Cloud

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[About Ruangguru](#)

With Google Cloud, Ruangguru has developed a range of apps that streamline and enhance the delivery of education services in Indonesia.



## About Ruangguru

Ruangguru provides education applications and services to tutors, teachers and students across Indonesia, including an online 'marketplace' that enables students to select skilled and empathetic tutors for private tuition.

Industries: Education

Location: Indonesia

**Google  
Cloud  
results**

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Ruangguru  
delivers

- Supported growth to accommodate 2.5 million students and 100,000 teachers and tutors
- Minimise outages and slowdowns that compromised students' experience with applications
- Achieved stability that convinced teachers and tutors to trust a digital education platform for the first time

For Indonesian education services provider Ruangguru, convincing students who are already accustomed to

range of applications and services available for gaining the confidence of teachers and tutors who may be using a digital platform for the first time.

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Compute Engine

(<https://cloud.google.com/compute/>)

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

digital technologies to use online education services is comparatively easy. However, the challenge lies in convincing teachers, tutors and education officials to transition from traditional manual practices to faster, more efficient online methods of performing their tasks.

For example, tutors can sign up to an online marketplace run by Ruangguru that enables students to review their credentials before deciding whether or not to engage them for private tuition. Ruangguru's first

product was a tutoring marketplace that allows students to check information such as educational background and certifications, as well as posting and checking reviews of tutors' performance.

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—Stephanie Hardjo,  
Business Development  
Manager, Ruangguru

Since 2014, the business has expanded its product suite to include an online service that enables students to undertake mock school or University tests in preparation for their real exams. Ruangguru has also added a service that enables students to chat with teachers using online text and audio to obtain answers to questions, and uploaded more than 2,000 videos that allow school students to prepare for exams in subjects such as maths, science and social sciences.

“Our mission is to provide high-quality, accessible education services to people across Indonesia,” says Stephanie Hardjo, Business Development Manager, Ruangguru.

“We have to assure stakeholders that moving to a digital environment is an easier and safer option

than manual, paper-based alternatives, particularly in terms of access to data and the ability to administer tests,” says Hardjo. “To convince them to do so, we need to run our applications and services on a stable platform.”

This requirement for stability encompasses high levels of availability to ensure Ruangguru applications and services are available when students, tutors, and other stakeholders need to access them. The business also needed the scalability to support spikes in usage that can see up to 10,000 users concurrently accessing Ruangguru’s products, without compromising performance.

Ruangguru had trialled running its service on public cloud platforms but decided to move to Google Cloud. Agung Firdaus, VP Engineering, Ruangguru, had experience with Google Cloud in previous roles and believed the platform would best meet Ruangguru’s requirements. The business then decided to migrate several of its applications to Google Cloud, and started re-architecting them to optimise their performance on the platform.

Ruangguru is now using Google App Engine to run most of these applications in a fully managed environment that takes care of all infrastructure requirements, and to scale to support demand peaks and ongoing growth. Other legacy applications are running primarily in Google Compute Engine virtual machine instances until the re-architecting process is complete. The business is using tools such as Google BigQuery and Google

Cloud Dataflow to warehouse and analyse big data in order to provide reports to stakeholders and partners. Google Cloud Pub/Sub sends messages between applications to enhance overall scalability.

## **Serving more users and accommodating more traffic**

Deploying Google Cloud has enabled Ruangguru to convince stakeholders of the merits and value of its platform in delivering education services. By mid-2017, Ruangguru had registered about 2.5 million students and 100,000 teachers and tutors. “Since we moved to Google Cloud, we gained the ability to serve more users and accommodate more traffic,” says Firdaus. “In addition, we can forge powerful partnerships with businesses to help promote and distribute our content. For example, we have created an official account in partnership with LINE, a messaging app popular with students, that blasts our content to a user base of 3.1 million students in Indonesia. Put simply, we can aim higher and engage existing and prospective customers more easily.”

Working with Google Cloud has enabled the education services provider to avoid the capital expenditure needed for on-premises or hosted servers, storage, and associated systems. Instead, Ruangguru can focus on the more efficient, budget-friendly approach of making regular payments for the resources it uses. Furthermore, the business only needs to allocate one engineer to system administration, with the remaining technology team

members focusing on product development and delivery.

The development team can easily replicate production environments for testing and staging, minimising the risks associated with pushing out updates and enhancements to its online products. These updates can be pushed out in minutes through an automated system, rather than in hours required when employing a manual process.

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—Agung Firdaus, VP  
Engineering, Ruangguru

## **Direct and indirect cost savings**

While the business has not fully calculated the financial benefits of using Google Cloud relative to a physical infrastructure, Firdaus believes the direct and opportunity cost savings are considerable. “As a start-up, we are growing and using a cloud platform saves us the administration required for on-premise or co-located physical servers,” he says.

Moving to Google Cloud has also enabled Ruangguru to eliminate outages that had compromised users’ experience of mock exams. “We’ve been able to reduce these errors and improve the user experience,” says Firdaus. “It’s easier to convince users to adopt other Ruangguru services now we have more stable products.”

For some teachers and tutors, Ruangguru represents their first experience with an online service for tests. “It’s very important that their first encounter with a digital application is error free so we can convince them this is something good and new,” says Hardjo. “That is one thing that Google Cloud has enabled us to achieve because it’s very stable and gives our users a better experience.”

Using Google Cloud has also enabled Ruangguru to maintain its leading position as a technology-enabled education provider in a highly competitive environment. “In Indonesia, we have many competitors that have been around for some time providing offline tuition,” says Hardjo. “They are now leveraging their established brands to move into online, so it’s becoming crowded. Our agility and digital experience is key to our success. Using Google Cloud is key to our value proposition.”